

TATA POWER SOLAR E-TORQUE

ETL DESIGN DOCUMENT

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1. Overview:

This document describes the design of ETL solution provided for this TATA POWER SOLAR E-TORQUE PROJECT. This document describes the requirement, data collection, data preparation, data ingestion, and data validation.

Build a scalable AWS Glue ETL pipeline to ingest transaction data from S3, transform it, and load it into Iceberg tables. The pipeline ensures accurate data processing, maximum bolt count selection, and proper history tracking for each location.

This pipeline is built using AWS Glue Visual ETL to process transaction data stored in S3. It transforms the data by creating location identifiers, filtering valid records, and storing both current and past records in the target table.

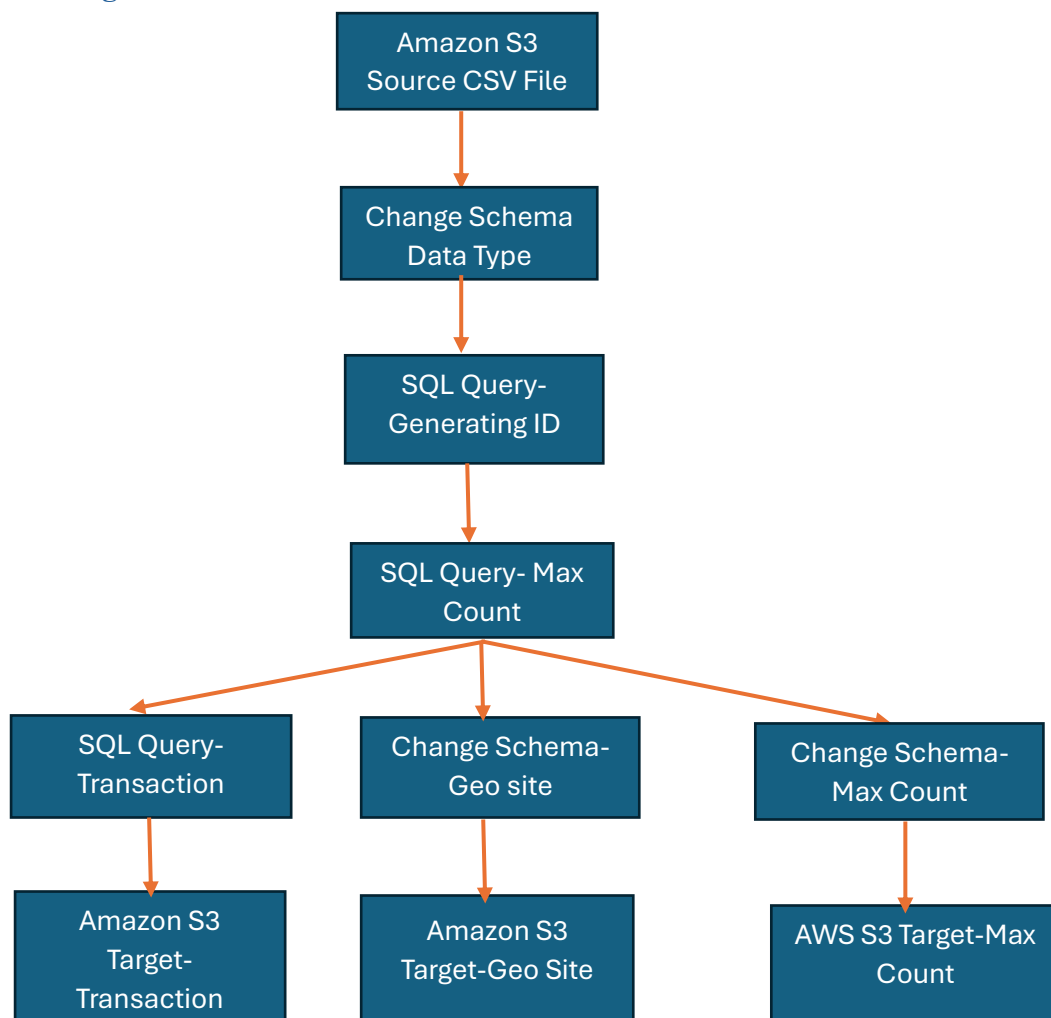
2.Design:

The pipeline follows a layered architecture with source, transformation, staging, and target layers. Data flows from S3 through schema transformation and SQL processing, then into staging tables before final loading into target tables.

2.1. Requirement:

- Site map Ingestion data from S3 into site map target table:
- Transaction Data Ingestion into Target table after Staging.
- Selective Transactional data Ingestion into the Final Target table.

2.2. ETL Design:



2.3.Source Details:

Attribute	Value
Source	Amazon S3
Path	s3://tatapowertorque2026/raw/Complete data/
Format	CSV
Schema	status, latitude, longitude, boltcount, modulecount, mmscount, torque, timestamp

2.4. Target Details:

Attribute	Value
Target	Aws Glue Data Catalog
Target table	Qa ods final
Storage Format	Iceberg
Schema	log_st, latitude_id, longitude_id, mms_count, module_count, bolt_count, torque, log timestamp, valid_start, valid end, ins_dt, upd_dt, latest_flag

3.Explanation of Each Block:

3.1.Amazon S3- Source CSV File:

- This Node reads the Source CSV File from S3 Bucket and passes the data to next step.

3.2.Change Schema:

- This Node changes the Data type of the Columns that are present in the CSV File.

3.3.SQL Query- Generating gps id:

- This node generates a **Deterministic GPS ID** using DENSE_RANK(). This ensures that the same coordinate pair always receives the same ID across different runs, maintaining data consistency without requiring a secondary master file.

3.4.SQL Query- Filter

- This node filters records where status = RUN and applies deduplication using ROW_NUMBER(). It selects the record with the highest bolt_count for each latitude and longitude.

3.5. Change Schema-Site:

- This node outputs gps_id,latitude_id,longitude_id and site_geo_location from SQL Query-Filter node.

3.6.AWS Data Glue catalog- Site Geo Target Table:

- This node is target node to store the Geo site data into the target table `geo_site_details_tb1`.

3.7.SQL Query—Transaction:

- **SCD Type 2 Implementation:** This node executes the "1-to-0 Flag Flip" to maintain a perfect historical audit trail.
- **Deactivation of Old Records:** It identifies existing active records (`latest_flag = 1`) for the same coordinates and "closes" them by setting `latest_flag = 0` and `valid_end = System Date - 1 (Yesterday)`.
- **Activation of New Records:** It inserts the fresh transaction as the new active record with `latest_flag = 1`, `valid_start = System Date (Today)`, and `valid_end = NULL`.
- **Gapless Timeline:** This ensures that the history is continuous—the previous state ends on the day before the new state is recorded in the warehouse.

3.8.AWS Data Glue Catalog- Transaction Target table:

- This node is a transaction target table, where store transaction data. It marks active record with 1 and history or old record with 0.

3.9. Change Schema- Max count

- This node output the row which has a maximum bolt count for each latitude and longitude pair.

3.10.AWS Data Glue Catalog—Staging Max count:

- This is the staging target table, which stores staging data. It contains latitude and longitude pairs with status RUN and the highest bolt count.

4.Business Logic:

4.1. Site Map Ingestion

- Read raw CSV data from S3.
- Convert raw string fields into proper types (double, int, timestamp).
- Generate deterministic GPS IDs using `DENSE_RANK()` so the same latitude/longitude pair always maps to the same ID.
- Create a `site_geo_location` string identifier for each coordinate pair.
- Store results in the `geo_site_details_tb1` table for consistent site mapping.

4.2. Transaction Data Ingestion (Staging)

- Filter records where `status = 'RUN'`.
- Apply **deduplication logic** using `ROW_NUMBER()` partitioned by latitude/longitude.
- Select the record with the **maximum bolt_count** for each coordinate pair.

- Store staging results in **qa_ods_staging** for intermediate validation.

4.3. Final Transaction Data (SCD Type 2)

- Implement **Slowly Changing Dimension (Type 2)** logic:
 - **Deactivate old records:** For coordinates already active (latest_flag = 1), set latest_flag = 0 and valid_end = yesterday.
 - **Activate new records:** Insert the new transaction with latest_flag = 1, valid_start = today, and valid_end = NULL.
- This ensures a **gapless timeline**: every location's history is continuous, with no missing days.
- Store results in qa_ods_final, which contains both active and historical records.

4.4. Audit & History Tracking

- Each record carries audit columns:
 - valid_start, valid_end → defines the active period.
 - ins_dt, upd_dt → timestamps for insertion/update.
 - latest_flag → indicates whether the record is current (1) or historical (0).

5. Job Bookmarking

- **Purpose:** Prevents reprocessing of the same data when the job runs multiple times. Ensures incremental ingestion rather than full reloads.
- **Configuration:**
 - **Bookmark option:** job-bookmark-enable
 - Location: Stored in Glue's metadata store.
- **Behavior:**
 - On first run: All data in the source path is processed.
 - On subsequent runs: Only new or changed files in the S3 source path are processed.
- **Business Benefit:**
 - Saves compute cost and time.
 - Ensures data freshness without duplication.
 - Supports continuous ingestion pipelines.

6. Testing the ETL Pipeline:

- Test the ETL pipeline, sample CSV files were placed in the S3 source location, and the pipeline was executed using AWS Glue Visual ETL.

- The pipeline first validated schema conversion and ensured only records with status RUN were processed further.
- It then verified that the maximum bolt_count record was selected correctly for each latitude and longitude combination.
- Existing records in the target table were checked to confirm they were updated as history when matching new records arrived.
- The latest records were inserted with correct values for valid_start, ins_dt, upd_dt, and latest_flag set to 1.
- Finally, the target table was reviewed in Athena to confirm that active and historical records were stored correctly after pipeline execution.

7.Security & Permissions

7.1.IAM Role

- Role Name: GlueS3ToAuroraRole
- Purpose: Provides Glue jobs with the necessary permissions to read from S3, write to S3, and update Glue Catalog tables.
- Action: Create a new IAM role specifically for this Glue job to ensure least-privilege access and separation of duties.

7.2. Required Policies

- **Amazon S3 Permissions:**
 - s3:GetObject → Read source files from S3.
 - s3:PutObject → Write transformed data to target S3 paths.
 - s3:DeleteObject → Manage staging/temporary files if needed.

8.Conclusion:

The AWS Glue pipeline delivers a scalable and reliable solution for ingesting site and transaction data from S3 into Iceberg tables. With schema transformations, deduplication, and SCD Type 2 logic, it ensures accurate processing, complete historical tracking, and efficient incremental ingestion for analytics and reporting.